Internal Research Brief: Multimodal RAG for Enterprise Support

This sample report is intentionally small. It gives the practical lab enough PDF text to demonstrate how a multimodal assistant can combine report paragraphs, chart images, and video transcript evidence.

Why text-only RAG misses important evidence

A normal text RAG system works well when the answer is present in paragraphs. It fails more often when the answer is embedded in a chart, architecture diagram, screenshot, or training video. In those cases, the retriever may find adjacent text but not the visual or spoken evidence that actually answers the question.

Recommended design pattern

For classroom and prototype systems, the recommended approach is caption-index plus late vision. Generate image captions once during ingestion, store the captions with metadata, and call a vision-capable model only when a retrieved image is central to the answer. This lowers cost while keeping the system explainable.

Source type	Searchable representation	Critical metadata
PDF text	Chunked page text	file name, page number, chunk id
Chart image	Vision-generated caption and structured facts	image path, chart title, source type
Video	Transcript chunk	video title, start time, end time

Evaluation checklist

A useful multimodal RAG evaluation set should include questions where the correct evidence comes from different modalities: one from PDF text, one from a chart, one from a video timestamp, and one requiring comparison across sources.

Implementation Notes for the Assistant

The assistant should preserve source traceability. Every retrieved object should carry source_type metadata so the final answer can say whether it used a PDF paragraph, chart image, or video segment.

The retriever should not send every image to a vision model during every query. A cheaper retrieval pass should shortlist evidence first. The expensive model call should happen late, only if visual evidence is relevant.

Practical success criteria

Criterion	Expected behavior
Modality recall	At least one relevant source from the correct modality appears in the top results.
Citation quality	Answers include PDF page, image filename, or video timestamp.
Cost control	Vision calls happen during captioning or late reasoning, not for every stored item on every query.